

Nicolás Violante

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EDUCATION

Inria, Université Côte d'Azur, Sophia Antipolis, France Oct. 2022 – Dec. 2025
Ph.D. in Computer Science. Advisor: George Drettakis

- Conducted research on Gaussian Splatting reconstruction and scene manipulation, training diffusion models for relighting and compositing.

École Normale Supérieure Paris-Saclay, Paris, France Oct. 2021 – Oct. 2022
M.Sc. in Mathematics, Vision, and Learning (with highest honours, GPA 17.38/20)

Universidad de la República, Montevideo, Uruguay Mar. 2015 – Oct. 2020
Electrical Engineering. (GPA 11/12)

EXPERIENCE

École Nationale des Ponts et Chaussées, Paris, France Feb. 2026 – Present
Postdoctoral Researcher. Mentor: Vincent Lepetit

- Conducting research on scene understanding and 3D representations

Adobe Research, San Francisco, United States Sept. 2024 – Nov. 2024
Research Intern. Mentor: Thibault Groueix

- Developed a 3D reconstruction method for repetitive elements for Gaussian Splatting, leading to a publication at SIGGRAPH 2025

Inria, Sophia Antipolis, France May 2022 – Sept. 2022
Research Intern. Mentor: George Drettakis

- Developed a camera pose estimation pipeline for single vehicle images to enable 3D GAN training and rendering with NeRFs, leading to a publication at Eurographics 2024

Digital Sense, Montevideo, Uruguay Oct. 2019 – Aug. 2021
R&D Engineer

- Implemented image enhancement techniques for large-scale (100 km²) remote sensing imagery
- Developed a workstation for semi-automatic industrial anomaly detection on leather samples

PUBLICATIONS

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- [1] A. Artykov, T. Ravaud, **N. Violante**, V. Lepetit, “*Articulation in Prime: Primitive-Based Articulated Object Understanding from a Single Casual Video*”, **NeurIPS**, 2026
 - [2] **N. Violante**, G. Kopanas, L. Franke, J. Philip, G. Drettakis, “*Lighting-Consistent Object Transfer Across Radiance Fields*”, **Computer Graphics Forum (EGSR)**, 2026
 - [3] **N. Violante**, A. Meuleman, A. Gauthier, F. Durand, T. Groueix, G. Drettakis, “*Splat and Replace: 3D Reconstruction with Repetitive Elements*”, **SIGGRAPH**, 2025
 - [4] **N. Violante**, A. Gauthier, S. Diolatzis, T. Leimkühler, G. Drettakis, “*Physically-based Lighting of 3D Generative Models of Cars*”, **Computer Graphics Forum (Eurographics)**, 2024

SKILLS

Programming: Python (Pytorch, OpenCV, Diffusers), C/C++, CUDA, Git, Slurm
3D & Rendering: SfM/COLMAP, 3DGS, NeRF, Physically-Based Rendering, Blender, Adobe Illustrator
Generative AI: Flow/Diffusion Models, GANs, VAEs
Languages: English, French, Spanish

AWARDS

Sophie Germain Scholarship, Fondation Mathématique Jacques Hadamard, 2021
Santander Iberoamérica Scholarship, 2019

SERVICE

Reviewer: NeurIPS, Eurographics, SIGGRAPH Asia, Computer Graphics Forum, Computers and Graphics.